

PROJECT TOPIC: INSTITUTIONAL GREEN AUDIT OF COLLEGE CAMPUS

Topic Description: Comprehensive Green Audit of Junior College Campus as per State Board Guidelines, Evaluating Energy Consumption, Water Management, Waste Segregation, Campus Biodiversity, and Carbon Sequestration

Educational institutions are micro-societies that consume substantial natural resources, generate daily municipal waste, and house hundreds of young citizens who will shape future environmental policy. A Junior College campus operates lighting fixtures, computer laboratories, science chemistry benches, administrative offices, and canteens. Without systematic environmental monitoring, campuses inadvertently contribute to resource wastage, high grid electricity consumption, and environmental degradation.

A **Green Audit** is an organized, comprehensive inspection and accounting process designed to assess an educational institution's environmental practices, resource use efficiency, and ecological footprint. Mandated by the National Assessment and Accreditation Council (NAAC) and recommended by the Maharashtra State Board curriculum, a Green Audit systematically evaluates five foundational environmental pillars: (1) Energy Conservation; (2) Water Security; (3) Solid & Hazardous Waste Management; (4) Campus Biodiversity & Green Cover; and (5) Carbon Footprint Mitigation.

This project conducts an empirical, student-led institutional Green Audit of our Junior College campus in Maharashtra. Guided by the textbook methodologies prescribed in Chapter 4 of the Standard 12 EVS curriculum, this study surveys campus infrastructure, quantifies resource flows, benchmarks sustainability indicators, and formulates an action plan. ¹ environmental master plan to transform our campus into a certified low-carbon green institution.

2. Importance of the topic (विषयाचे महत्त्व)

Executing an institutional green audit is of paramount educational, economic, and ecological significance:

Key Pillars of Campus Green Audit Importance :-

- **Transforming Theory into Lived Conservation Practice:** Bridges the gap between abstract textbook environmental concepts and tangible physical campus reality, empowering students to act as practical environmental auditors within their own school.
- **Substantial Institutional Cost Reduction:** Identifying undetected water pipeline leaks, inefficient lighting, and unmonitored computer lab loads saves tens of thousands of rupees annually on institutional utility bills.
- **Ensuring Statutory Compliance with Waste & Safety Rules:** Auditing ensures that hazardous laboratory chemicals, biological waste, and electronic waste (e-waste) are disposed of strictly in accordance with statutory national environmental regulations.
- **Fostering Campus Biodiversity Sanctuaries:** Promotes the mapping, labelling, and scientific expansion of native tree species, transforming urban college campuses into vital biodiversity stepping-stone habitats for birds and pollinators.

A green campus serves as a living, functioning model of sustainable development for the surrounding civic community.

3. Objectives of project work (प्रकल्प कार्य उद्दिष्टे)

The institutional green audit was governed by structured, measurable environmental auditing objectives:

Structured Project Objectives :-

- **Campus Energy Consumption Audit:** To inventory all lighting, fans, computer systems, and lab equipment across 24 classrooms and offices, calculating total daily kilowatt-hour (kWh) demand.
- **Water Management & Harvesting Survey:** To audit daily water consumption, inspect plumbing fixtures for leakage losses, and assess the catchment potential of campus rooftop rainwater harvesting.
- **Waste Generation & Segregation Assessment:** To quantify daily solid waste generated across classrooms, canteens, and science labs, assessing the percentage of waste segregated at source.
- **Floral Biodiversity & Tree Inventory:** To map, count, and taxonomically identify all tree species on campus, calculating their canopy cover and annual carbon sequestration capacity.
- **Formulating the 'Green Campus Action Charter':** To develop a student-led, 5-point institutional charter for achieving zero-waste status and expanding solar rooftop generation.

The audit delivers an authentic empirical evaluation of institutional sustainability.

4. Project Work Methodology (प्रकल्प कार्यपद्धति)

A rigorous physical inspection, utility billing review, and botanical inventory methodology was executed:

Methodological Phases Followed :-

- **Phase 1 – Audit Team Formation & Questionnaire:** Formulated a 5-member student green audit committee under the guidance of our EVS faculty, designing customized physical auditing checklists for each campus block.
- **Phase 2 – Energy & Electrical Inventory:** Inspected all 24 classrooms, 4 science labs, the library, and administrative offices; logged power ratings (Watts) and average operating hours of every fixture.
- **Phase 3 – Water Flow & Plumbing Audit:** Measured water flow rates from washroom taps using graduated measuring beakers; conducted a 24-hour visual leak audit of all overhead storage tanks and pipeline joints.
- **Phase 4 – Daily Waste Sorting & Weighing:** Collected and sorted campus waste over 3 consecutive weekdays using digital hanging scales to categorize paper, wet food scraps, and plastic wrappers.
- **Phase 5 – Botanical Tree Mapping:** Surveyed the campus perimeter to measure tree girths at breast height (GBH) and identify species using regional botanical keys and mobile plant-identification tools.

All recorded figures represent direct physical measurements verified across institutional administrative records.

5. Observations (निरीक्षणे)

Comprehensive physical auditing of the Junior College campus (1,250 students and staff; 3.2 acres) yielded rich empirical findings across all 5 green domains:

5.1 Institutional Green Audit Matrix (Baseline Performance & Resource Inefficiencies)

Audit Domain	Current Baseline Infrastructure & Consumption	Identified Inefficiencies & Ecological Findings
1. Energy Audit	- Total Connected Load: 38.4 kW - Average Monthly Consumption: 3,450 kWh - Fixtures: 140 fluorescent tubes (40W), 85 ceiling fans (75W), 45 computers.	Over 65% of classrooms still use obsolete 40W fluorescent tubelights rather than 18W LEDs; computers left running in standby mode overnight accounted for 180 kWh/month of pure energy waste.
2. Water Audit	- Daily Water Consumption: 18,500 Litres/day (Municipal + 1 Borewell) - Washroom fixtures: 32 taps, 12 flush tanks.	Auditing revealed 4 leaking washbasin taps dripping ~12 Litres/hour each (losing 1,150 Litres of drinking water daily); rooftop rainwater harvesting pipelines were clogged with silt.
3. Waste Audit	- Total Daily Waste: 48 kg/day - Paper: 22 kg (45.8%); Canteen Food: 18 kg (37.5%); Single-use plastic wrappers: 8 kg (16.7%).	Only 22% of classrooms maintained segregated waste bins; unsegregated plastic wrappers and canteen wet waste were being collected together and handed over to municipal trucks.
4. Campus Biodiversity	- Total Trees Counted: 68 mature trees - Dominant Species: Neem (18), Gulmohar (14), Ashoka (12), Peepal (8), Banyan (4), Rain Tree (12).	Campus tree canopy covers approximately 22% of total open ground; provides nesting habitat for 14 bird species; estimated carbon sequestration: 1.45 tonnes CO₂/year .
5. Carbon Footprint	- Annual Grid Emissions 5 × 12 × 0.82 = 33.95 tonnes CO₂e - Staff & Student Commuting: 18.4	Grid electricity accounts for 64.8% of total institutional carbon emissions; high reliance on motorized two-

6. Analysis of data (माहितीचे विश्लेषण)

Analytical evaluation of the campus green audit data reveals critical high-leverage intervention points for institutional decarbonization:

Core Analytical Inferences :-

- **Low-Cost Fixes Deliver Massive Resource Savings:** The water audit proved that minor, neglected maintenance flaws cause massive cumulative waste. Four leaking washers (costing under ₹100 to repair) wasted enough water to supply 8 families for an entire year. Preventive plumbing audits should be conducted monthly.
- **Rooftop Solar Feasibility:** The college building features 1,100 square meters of unshaded rooftop concrete terrace. Installing a 25 kW grid-tied rooftop solar photovoltaic (PV) system would generate ~36,000 kWh annually, covering over 85% of institutional electricity needs, reducing campus carbon emissions by 29.5 tonnes CO₂ annually, and achieving capital payback within 4.2 years.
- **The Untapped Circularity of Campus Paper Waste:** Daily academic operations generate 22 kg of dry paper waste (exam papers, rough drafts, circulars). Establishing an institutional partnership with a local paper recycling mill will convert waste paper into recycled drawing sheets and certificates, realizing a closed-loop circular economy.
- **Ecological Value of Native Canopy:** While ornamental exotic trees like Ashoka provide visual aesthetics, they support negligible insect and avian life. Future campus plantation drives must focus strictly on native fruit-bearing species (Jamun, Bael, Fig, Amla) to enhance urban biodiversity corridors.

The analysis proves that a college green audit converts educational campuses into living laboratories of climate resilience.

7. Results and Conclusions (निष्कर्ष)

Conclusion: Conducting an institutional Green Audit demonstrates that sustainable campus management is both ecologically transformative and financially profitable. By repairing plumbing leaks, transitioning to energy-efficient LED fixtures, composting canteen organic waste on-site, and nurturing native biodiversity, our college can drastically slash its carbon footprint and inspire students to become proactive climate champions.

The Student Green Campus Action Charter (5 Directives) :-

- ❑ Mandate that the last student leaving any classroom switches off all lighting and ceiling fans.
- ❑ Replace all remaining fluorescent tubelights with energy-efficient 18W LED fixtures over the next term.
- ❑ Implement strict 2-bin segregation (Green for Canteen Wet Waste, Blue for Paper) in every corridor.
- ❑ Revive the rooftop rainwater harvesting system to recharge the campus groundwater borewell.
- ❑ Plant 25 native medicinal and fruit-bearing trees along campus boundaries during Van Mahotsav.

8. Bibliography / References (संदर्भ सूची)

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